

PULSE — Regime-Conditional Crude Spread Engine

Methodology & Walk-Forward Results · Generated 2026-06-22 06:54 UTC · Reproduce: `python -m backend.research.walkforward` · `python -m backend.research.methodology_pdf`

1. Problem

Identify when crude calendar spreads (front carry, mid-curve carry, front butterfly) deviate from their *regime-conditional* fair value. Output a single ranked opportunity per day with full receipts: fair, 80% band, residual z-score, top drivers, historical analogs.

2. Universe

6 instruments — 3 Brent (M1-M2, M3-M6, front fly M1-2×M2+M3) + 3 WTI mirror instruments. Brent legs from /Data C1-C31 daily settlements (2016-2026, 2,712 trading days). WTI legs from synthesised daily settlements (last 1-min mid per session, 2021-2026, 1,676 trading days — flagged ESTIMATE pending the real exchange file).

3. Regime grids — composite + pooled

Composite (Sprint 3): 27 cells = **CURVE** {CONTANGO / NEUTRAL / BACK on Brent M1-M12 at -\$2, +\$5} × **INVENTORY** {LOW / AVG / HIGH on US crude stocks vs 5y seasonal at ±4%} × **VOL** {CALM / NORMAL / STRESSED on Brent 20d realised vol at 20%, 35%}. Hard thresholds — mentor can answer ‘why this regime?’ in one line. ~10-15 cells/spread have usable n ; the rest are economically implausible.

Pooled (Phase 2.5): 3 cells/spread = curve axis only {CONTANGO / NEUTRAL / BACK}. Same thresholds; inventory + vol still surface on the UI as regime context but no longer fragment the training set. Average ~770 rows per cell (vs ~150 composite) — explicitly addresses the Sprint 4 finding that per-cell data was too thin to beat a 252-d rolling z .

4. Feature set (Phase 2.8.2 expansion)

20 Brent features per cell (was 11): *Curve shape* — M1-M12, its square, and *curvature* (M1 – 2×M6 + M12) for fly-specific information. *Front state* — Brent close + 5d return + 20d realised vol. *Inventory* — US crude vs 5y seasonal AND **inventory surprise** (weekly Δ minus its 4-week MA). *Seasonality* — sin/cos of day-of-year. *Lagged spreads* — M1-M2 / M3-M6 / fly lag-1.

Phase 2.8.2 alpha features (new) — **COT managed-money percentile** (rolling 156-week, contrarian crowding), **3-2-1 crack + gasoline crack** (refining-margin pressure on the front), **WTI-Brent spread** (Atlantic-basin arb), **real rate** (DGS10 – 5y TIPS breakeven; storage-cost driver), **OVX/VIX ratio** (crude vs equity vol risk-premium), and **days to front-month expiry**. WTI cells append 5 more (WTI close, M1-M12, 3 lag-1 spreads). The target spread’s own lag is removed to avoid self-leakage.

5. Per-cell model competition

We *do not* pick a single regression family a priori. For each (spread, regime) cell, **seven candidates** compete on 5-fold TimeSeriesSplit cross-validated R^2 (Phase 2.8.1 expansion):

Linear: Ridge (shrinks coefficients — good with correlated predictors), Lasso (zeroes out noise features — good when 3-4 signals dominate), ElasticNet (L1+L2 blend — mixed correlation), Huber (robust loss — COVID-2020 / Russia-2022 outliers).

Gradient boosting (Phase 2.8.1): XGBoost, LightGBM, CatBoost — shallow trees with strong regularisation that capture non-linear interactions between features. Win in cells where linear leaves alpha on the table.

Winner = max mean CV R^2 , simplicity tiebreak within 0.005: linear families rank above boosters (ElasticNet > Lasso > Ridge > Huber > XGBoost > LightGBM > CatBoost), so interpretability wins when CV scores are close. Quantile regressors (p10/p50/p90) are fit separately for the 80% confidence band.

6. Trading rule

At each daily settle: classify today’s composite regime → look up winner model for (spread, regime) → predict fair, p10, p50, p90 → $z = (\text{actual} - \text{fair}) / \text{residual}_\sigma$ from training. Direction = SELL if $z > +0.5\sigma$, BUY if $z < -0.5\sigma$, else NEUTRAL. Target = p50 (mean reversion). Stop = entry $\pm 1.5\sigma$. Horizon = 20 trading days.

7. Walk-forward design

Refits every quarter from 2024-Q1 through 2026-Q2 (34 refits). Each refit re-runs the full 4-model competition on data $\leq$ cutoff, then we walk forward day-by-day to the next cutoff, generating signals. **Phase 2.5 runs BOTH grids in parallel** (composite + pooled) over the same refit dates and the same trading rule, then compares each to a regime-*unaware* 252-day rolling z -score baseline — isolating the contribution of regime conditioning and of grid resolution.

8. Phase 2.6 — gated blend (production rule)

Phase 2.5 surfaced the BACK × {Lasso, Huber} slice as the only (spread × regime × winner) subset where the pooled engine demonstrably beats baseline in walk-forward. **Phase 2.6 codifies that finding as a production rule** and tests it as a third leg alongside composite / pooled / baseline. For each (date, spread): (i) if the pooled candidate has *regime_pooled*=‘BACK’, *winner_model* ∈ {Lasso, Huber}, AND $|z| \geq 0.5\sigma$ → take the pooled signal; (ii) else fall back to the 252-day rolling- z baseline. This routing is the exact rule shipped under PULSE_GATED_BLEND=1, so the table on page 2 measures the strategy a trader would actually run.

9. Phase 2.7 — position sizing for the regime leg

Phase 2.6 left max drawdown on the gated blend at -271 (vs -169 baseline) — the cost of concentrating 97 high-Sharpe regime fires on a small number of days. **Phase 2.7 scales the regime-leg notional** while leaving the baseline leg at 1.0, asking whether sizing compresses the DD without giving up the +1.332 regime-leg Sharpe. Three modes simulated end-to-end as a fourth walk-forward leg behind PULSE_GATED_SIZE=<mode>:

full — scale 1.0 (sanity check; identical to gated_blend). **half** — scale 0.5 (uniform risk reduction). **kelly** — per-spread Kelly fraction $f^* = p - (1-p)/b$ computed at each refit boundary from prior closed regime-leg PnLs (p = win rate, b = mean_win / |mean_loss|). Clamped to [0.10, 1.00]; defaults to 0.50 when fewer than 5 prior trades exist. The schedule is expanding-window so the per-spread sample grows over time. Live inference reads the per-spread Kelly at the latest refit boundary from sized_blend_summary.kelly_per_spread_latest.

10. Phase 2.8 — real model + real features

Phase 2.8.1 adds **XGBoost / LightGBM / CatBoost** to the per-cell competition so cells with non-linear interactions aren't forced to fit a linear model. Boosters fit only on cells with ≥ 80 train rows (smaller samples can't reliably distinguish trees from linears via CV); shallow trees (depth 3-4) and low learning rate (0.05) keep them from memorising. Boosters rank below linears in the tiebreak so interpretability still wins on ties.

Phase 2.8.2 nearly **doubles the feature set** (11 $\rightarrow$ 20 Brent features) with the alpha-bearing predictors documented in §4: COT crowding, inventory surprise, curvature, refining cracks, real rate, OVX/VIX ratio, WTI-Brent arb, days-to-expiry. The aim is to give every model family — linear and tree — a richer set of predictors so the gated-blend Sharpe lifts by tightening the pooled BACK fair-value estimates the gate rule relies on.

Phase 2.8.3 widened the gate to admit boosters (GATED_WINNERS = {Lasso, Huber, XGBoost, LightGBM, CatBoost}). The widened gate is methodologically correct — the regime leg's reported Sharpe lifted from +0.369 to **+0.888** across 244 fires (was 71) — but the headline blended Sharpe stayed flat (+0.389 $\rightarrow$ +0.384) because most of the newly-routed booster trades had baseline-z direction matching pooled-z direction. The widening REASSIGNS attribution rather than CREATING new alpha at the blended level.

Phase 2.8.4 — **global model with regime-as-feature**. The 2018-2026 walk-forward (Phase 2.8.8) showed the regime-unaware baseline beats every per-cell variant on NET Sharpe — so the per-cell *split* is itself the problem. Phase 2.8.4 collapses the grid: **ONE** model per spread trained on *all* rows, with the composite regime fed as **9 one-hot axis columns** (curve / inventory / vol — 3+3+3) instead of partitioning the data. Same 7-model competition. Each spread now trains on $\sim 5\times$ more rows per refit, giving boosters a real shot at picking up regime $\times$ feature interactions the per-cell linear winners can't express. Reported alongside the other modes on page 2 — same NET cost model, same horizon, same 34 refits.

10b. Phase 2.8.5 — soft regime probabilities

Complementary test to 2.8.4. The hard pooled grid forces **discontinuous** regime assignment at -\$2 and +\$5 on M1-M12; two days at -1.9 and -2.1 flip cell, model, and residual scale even though the market is virtually identical. Phase 2.8.5 replaces the indicator with a **logistic membership function** at each threshold (bandwidth \$1/bbl for curve and inventory, 2.5pp for vol) — at the boundary a day is 50/50 across adjacent regimes, far from the boundary it collapses back to the hard label. Each day's prediction is the *posterior-weighted blend* across all trained cells for that spread (point, quantiles, and residual variance all blended). Bandwidths come from the trader thresholds, not tuned to PnL.

Honest framing: hard pooled recovers as bandwidth $\rightarrow 0$; a flat prior across cells recovers as bandwidth $\rightarrow \infty$, which is approximately the 2.8.4 global model. So 2.8.5 sits on the interpolation between 'hard split' and 'single global model' — the question it answers is whether a *small* amount of cell-sharing recovers alpha the discontinuous cut leaks. Reported on page 2 alongside the other modes under the same 34-refit walk-forward and the same NET cost model.

11. Phase 2.8.6 — transaction costs

Every Sharpe/PnL number above is **gross** — per-trade transaction cost is not modelled by the walk-forward's spread-move PnL. Phase 2.8.6 layers a defensible cost model on top of the existing trade tape so the table on page 2 reports **NET (after-cost)** Sharpe alongside gross.

Cost model. Per-leg, per-side: \$0.0025/bbl commission + clearing + brokerage ($\sim$ \$2.45 per 1,000-bbl contract) + half bid-ask slippage (\$0.0050/bbl for front M1/M2, \$0.0075/bbl for deferred M3-M6). Round-trip cost = $N \text{ legs} \times 2 \text{ sides} \times (\text{commission} + \text{half-spread})$. That gives: **2-leg M1-M2 = \$0.030/bbl RT, 2-leg M3-M6 = \$0.040, 3-leg fly = \$0.050**. Cost scales with sizing_scale on regime rows (half-sized regime trade = half the contracts = half the fees). NEUTRAL trades incur zero cost (no fill).

Cost is applied at the aggregation step only — per-cell CV is on gross fair-value R^2 , so winner selection is unaffected and no retraining is required. Live inference does *not* subtract cost at entry; the cost-aware NET metric exists purely for backtest reporting.

Walk-forward results — 2024-2026

Refits: 34 · Records/mode: 8,421 · Horizon: 20 trading days · Universe: 6 spreads · Composite grid: 27 cells · Pooled grid: 3 curve cells · Gated blend (Phase 2.6): regime engine fires on BACK + {Lasso,Huber} + $|z| \geq 0.5\sigma$, else 252d z baseline (regime/baseline split: 1,567 / 8,379) · Sized blend (Phase 2.7): regime-leg notional scaled by {full=1.0, half=0.5, kelly=per-spread f^* }

Overall — composite / pooled / gated blend / baseline

Metric	Composite (27)	Pooled (3)	Gated blend	Baseline 252d z	Δ gated vs base
Signals fired	6380	7160	6598	6140	—
Hit rate	57.9%	59.4%	63.8%	65.2%	-1.4%
Mean PnL (\$/bbl)	+0.133	+0.142	+0.148	+0.177	-0.029
Median PnL	+0.070	+0.080	+0.130	+0.150	—
Total PnL	+846.2	+1013.5	+974.4	+1087.1	—
Sharpe (ann.)	+0.38	+0.41	+0.41	+0.48	-0.07
Max drawdown	-102.1	-125.3	-275.7	-168.9	—

Phase 2.6 finding: even the gated blend can't pull the headline above baseline — the BACK + {Lasso,Huber} slice was strong in Phase 2.5 but it didn't survive a clean walk-forward of the gating rule itself. Sharpe (ann.): composite +0.38, pooled +0.41, **gated +0.41**, baseline +0.48. Gate routing: 1,567 regime fires / 8,379 baseline fallbacks (15.8% of records hit the engine). The gate doesn't beat baseline in walk-forward. Next: try a wider feature set inside the pooled regression (EIA inventory surprise, COT positioning, OVX, term-structure derivatives), or revisit gating criteria — the Phase 2.5 slice may have been window-specific.

Per-spread Sharpe — composite / pooled / gated / baseline (Δ vs baseline)

Spread	Comp Shp	Pool Shp	Gate Sig	Gate Shp	Base Shp	Δ Gate
brent_fly_123	+1.11	+1.02	1106	+1.28	+1.52	-0.24
brent_m1_m2	+0.70	+0.61	1490	+0.71	+0.77	-0.06
brent_m3_m6	-0.15	+0.05	1619	+0.09	+0.06	+0.02
wti_fly_123	+0.76	+0.91	731	+0.86	+0.97	-0.11
wti_m1_m2	+0.54	+0.33	817	+0.33	+0.33	+0.00
wti_m3_m6	+0.31	+0.36	835	+0.10	+0.39	-0.29

Gated-blend slices — which leg fired (regime vs baseline) and winner mix

Gated-blend slice	n	Hit	μ PnL	Sharpe
source=baseline	5071	64.7%	+0.137	+0.38
source=regime	1527	60.7%	+0.184	+0.48
winner=CatBoost	584	82.9%	+0.346	+1.21
winner=Huber	1788	58.7%	+0.086	+0.32
winner=UNKNOWN	107	80.4%	+0.334	+3.26
winner=XGBoost	1636	60.8%	+0.147	+0.47
winner=Ridge	2483	64.2%	+0.138	+0.30

Phase 2.7 — sized regime leg vs gated full / baseline

Metric	Gated full	Sized 0.5×	Sized Kelly	Baseline 252d z	Δ best vs base
Signals fired	6598	6598	6598	6140	—
Hit rate	63.8%	63.8%	63.8%	65.2%	—

Mean PnL (\$/bbl)	+0.148	+0.126	+0.123	+0.177	—
Total PnL	+974.4	+833.8	+814.6	+1087.1	—
Sharpe (ann.)	+0.41	+0.39	+0.38	+0.48	-0.07
Max drawdown	-275.7	-295.2	-295.1	-168.9	-106.9
Mean regime scale	1.0	+0.500	+0.342	—	—

Phase 2.7 finding: no sizing mode caught baseline at the headline Sharpe (best gated full +0.41 vs baseline +0.48). Max DD on the best variant: -275.7 vs baseline -168.9 (gated full: -275.7). Latest per-spread Kelly fractions: brent_m1_m2=0.18, brent_m3_m6=0.10, brent_fly_123=0.37, wti_m1_m2=0.44, wti_m3_m6=0.10, wti_fly_123=0.51.

Phase 2.8.6 — NET headline (after transaction costs)

Cost model: per-leg per-side \$0.0025 commission + half-spread slippage (\$0.0050 front / \$0.0075 deferred). RT cost per spread: M1-M2 \$0.030, M3-M6 \$0.040, fly \$0.050.

Mode	Gross / Net Sharpe	Gross / Net μ PnL	Mean cost / fire	n fired
Baseline 252d z	+0.48 / +0.37	+0.177 / +0.138	+0.0389	6140
Pooled (un-gated)	+0.41 / +0.29	+0.142 / +0.101	+0.0404	7160
Gated blend (2.8.3)	+0.41 / +0.30	+0.148 / +0.108	+0.0393	6598
Global (Phase 2.8.4)	+0.49 / +0.38	+0.177 / +0.136	+0.0410	6263
Pooled-soft (2.8.5)	+0.42 / +0.30	+0.141 / +0.101	+0.0404	7310
Sized half	+0.39 / +0.28	+0.126 / +0.092	+0.0343	6598
Sized kelly	+0.38 / +0.28	+0.123 / +0.090	+0.0329	6598

Spread	Gross Shp (gated)	Net Shp (gated)	Net Shp (base)	Δ Net vs base
brent_fly_123	+1.28	+1.02	+1.26	-0.24
brent_m1_m2	+0.71	+0.61	+0.67	-0.06
brent_m3_m6	+0.09	-0.01	-0.04	+0.03
wti_fly_123	+0.86	+0.67	+0.80	-0.13
wti_m1_m2	+0.33	+0.26	+0.26	+0.00
wti_m3_m6	+0.10	+0.02	+0.31	-0.29

Phase 2.8.6 finding: transaction costs erode Sharpe roughly uniformly across modes — baseline drag +0.10, pooled +0.12, gated_blend +0.11. Mean cost per fire sits at \$0.039-\$0.045/bbl depending on spread mix (boosters fire more on the fly which is the most expensive class). **Best net Sharpe: baseline at +0.372.** Gated blend net Sharpe (+0.298) vs baseline net (+0.372) = Δ -0.074 — the gated blend loses to baseline at the net headline. Transaction costs widen the gap because the regime fires concentrate on the fly (highest per-trade cost class). **Phase 2.8.4 (global model):** net Sharpe +0.380 — ties baseline (+0.372). Regime-as-feature is not a headline lift on this universe, but it doesn't cost anything either — the per-cell *split* was the binding constraint. **Phase 2.8.5 (soft pooled):** net Sharpe +0.297 — softening the curve thresholds does not lift pooled past baseline (+0.372). Discontinuity at the trader cuts isn't the binding constraint on this universe (mean modal weight 0.86, ~2.5 cells blended per fire).

Phase 5 — robustness: multi-horizon sweep + feature selection

Multi-horizon sweep (2.8.7). The walk-forward fixes a 20-trading-day forward horizon. But the models predict the *contemporaneous fair value* of the spread, so the entry signal is horizon-independent — only the hold/exit horizon changes realised PnL. We recompute every fired signal's PnL at $H \in \{5d, 10d, 20d, 30d\}$ over the validated baseline and global trade tapes (no retraining), NET of the same per-spread round-trip cost (horizon-independent — one round trip), Sharpe annualised $\sqrt{(252/H)}$.

Horizon	Baseline NET Sharpe	Global NET Sharpe
5d	+0.168	+0.395
10d	+0.306	+0.451
20d	+0.372	+0.380

30d	+0.530	+0.297
Spread	Baseline best H (NET)	Global best H (NET)
brent_fly_123	30d (+1.44)	10d (+1.06)
brent_m1_m2	30d (+0.73)	5d (+0.76)
brent_m3_m6	30d (+0.09)	10d (+0.36)
wti_fly_123	30d (+0.91)	20d (+0.59)
wti_m1_m2	30d (+0.69)	20d (+0.46)
wti_m3_m6	30d (+0.55)	5d (+0.60)

Horizon finding: the two engines' edges live at different horizons. The regime-unaware **baseline reverts slowly** — its NET Sharpe climbs monotonically with the hold and peaks at **30d** (+0.530 vs +0.372 at 20d); every spread's baseline edge peaks at the 30d bound. The **global regime model carries a faster signal** — its NET Sharpe peaks around **10d** (+0.451 vs +0.380 at 20d) and the front spreads (brent M1-M2) peak at 5d, decaying thereafter. This corroborates the live 30-trading-day time-stop on the baseline-led book and says the regime overlay, where used, should be harvested faster. **Caveat:** daily fills overlap, and longer horizons overlap more, so the absolute 30d Sharpe is the most overlap-inflated — read the ranking across H, not the level. It does not overturn the headline (baseline still wins at every common horizon).

Feature selection (stability selection). The global leg trains on all 22 (Brent) / 24 (WTI) base features + 9 regime one-hots. We pick a leaner set with Meinshausen-Bühlmann stability selection: a scaled Lasso is fit per spread at every refit cutoff and a base feature is kept if its non-zero-coef selection frequency across the 34 refits is $\geq 50\%$ (floor: top-6 by mean |coef|); the 9 regime one-hots are always retained. The lean per-spread sets are then re-run through the global walk-forward.

Spread	Feats kept	Lean NET	Full NET	Baseline NET
brent_fly_123	8/19	+0.97	+0.92	+1.26
brent_m1_m2	18/19	+0.43	+0.24	+0.67
brent_m3_m6	19/19	+0.19	+0.25	-0.04
wti_fly_123	14/24	+0.65	+0.59	+0.80
wti_m1_m2	21/24	+0.31	+0.46	+0.26
wti_m3_m6	17/24	-0.05	+0.32	+0.31

Feature-selection finding: a leaner set does **not** hold the NET headline. Lean-global NET Sharpe +0.308 trails both full-global +0.380 and baseline +0.372 — pruning costs ~ 0.07 , driven by the WTI M-spreads where the dropped features were load-bearing. It is, however, a real **interpretability** win where the edge is concentrated: the butterfly spreads prune hardest (brent fly keeps just 8/19 features — curvature, calendar, lags, COT and cracks), and lean even *beats* full-global on both flies and brent M1-M2. Verdict, in the Phase 2.8.x tradition: stability selection at the 50% threshold trades a small NET loss for interpretability rather than buying a free lunch; the full feature set stays the recommended default.

Phase 6 — data-driven regimes (HMM / change-point, 2.8.9)

The Phase 2 grid splits the curve axis on **hard trader thresholds** ($\text{CONTANGO} \leq -\$2 < \text{NEUTRAL} \leq +\$5 < \text{BACK}$). This leg asks whether a *fitted* regime detector beats both the regime-unaware baseline and the hard-threshold pooled / global variants. We fit a **Gaussian mixture + causal sticky-HMM** over the curve level (M1-M12) and its 5-day change, relabel the states ordinally by curve level (R0 = deepest contango), and re-run the same 7-model per-cell competition over the discovered regimes — same 20-day horizon, same NET cost model, same 34 refits. The detector is fit on each refit's training slice and the forward window is labelled causally (look-ahead-free).

What the data wants vs the trader thresholds. Averaged across refits, the headline K=3 detector places its curve cut points at $-\$0.81 / +\3.34 — versus the hard $-\$2 / +\5 . The data prefers a higher contango boundary than the trader's $-\$2$.

States (K)	Boundaries (\$)	Gross Sharpe	NET Sharpe	n signals
K=2	+2.00	+0.400	+0.279	6875
K=3 (headline)	-0.81 / +3.34	+0.410	+0.289	6882
K=4	-1.99 / +1.17 / +5.54	+0.322	+0.201	6976

Strategy	NET Sharpe
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baseline (regime-unaware 252d z)	+0.372
pooled (hard 3-bucket curve)	+0.293
global (regime-as-feature)	+0.380
HMM (data-driven K=3)	+0.289

Verdict (graded, Phase 2.8.x tradition): the data-driven detector does not beat the regime-unaware baseline (+0.289 vs +0.372) and ties/trails the hard-threshold pooled grid (+0.293). Learning the regime boundaries from the data does not unseat the headline — consistent with 2.8.4/2.8.5, the regime *partition* is not the binding constraint on this 6-spread universe. The honest read either way: the detector relocates the contango cut from the trader's -\$2 toward zero, which is a defensible interpretability result regardless of the NET ranking. State count K is a sensitivity check, not a tuned hyper-parameter — K=3 is fixed to match the 3 hard buckets.

Phase 7 — portfolio vol-targeting (2.8.10)

The last open Phase 2.8.x leg sizes the *book* for roughly constant risk, reusing two shipped primitives: `shock_engine.risk_scale` (per-position vol-target × stress de-risk) and `gated_select.select_decorrelated` (the desk's decorrelated book), plus a **portfolio overlay** that scales the whole book to a target vol from the trailing correlation matrix. Pure post-processing over the gated trade tape (same fired signals, reweighted notionals; no retrain). Books are normalised to mean notional 1.0 so max-drawdown is comparable; Sharpe and Calmar are leverage-invariant regardless. The four layers are toggled into ablation variants so each one's marginal effect is visible.

Book	Held	NET Sharpe	Max DD	Calmar
baseline (un-targeted)	6140	+0.372	-170.1	+4.99
gated (un-targeted)	6598	+0.298	-280.9	+2.55
+ decorrelation	4834	+0.251	-214.7	+1.92
+ risk parity (vol_scale)	4834	+0.235	-140.9	+1.87
+ stress de-risk	4834	+0.203	-141.0	+1.52
+ portfolio overlay (headline)	4834	+0.198	-112.0	+2.01

Verdict (graded): vol-targeting is a **drawdown-management tool, not an alpha tool**. The full overlay cuts the gated book's max-drawdown from -280.9 to -112.0 (−60%), but NET Sharpe falls +0.298 → +0.198 and Calmar +2.55 → +2.01 — the drawdown reduction does not pay for the return given up, so on risk-adjusted return it does not beat the un-targeted gated book, and the regime-unaware baseline (Calmar +4.99) remains the headline. The ablation localises the effect: per-position **risk parity** is the biggest single drawdown-reducer (it caps the high-\$-vol fly), and the **portfolio overlay** is the only layer that improves Calmar at the margin (it adds exposure to thin/hedged books and trims full/correlated ones). Consistent with the whole Phase 2.8.x arc: the machinery manages risk well but does not lift the NET-Sharpe headline on this 6-spread universe. For a drawdown-constrained mandate the overlay is worth the Sharpe; for a Sharpe-max mandate the baseline wins. **Caveat:** the stress detector is fit 2016–19 (OOS for 2020+; the 2018–19 tape rows see an in-sample stress read).

Phase 8 — per-spread gate (2026-06-22)

The Phase 2.6 production gate is a *single* rule applied uniformly to every spread (take the regime signal iff `regime_pooled==BACK` and the winner is gate-eligible and $|z| \geq 0.5$). But the per-spread lift table shows regime conditioning helps **unevenly**: the regime leg's OOS NET Sharpe beats baseline on the WTI front spreads but loses on every Brent spread and the M3-M6 carries. This phase replaces the uniform gate with a **per-spread enable decision** — the regime leg fires for a spread only when its OOS NET Sharpe beat baseline for that spread before each refit cutoff ($\geq$ min-sample evidence). Decided walk-forward (prior closed trades only, the Kelly-by-cutoff pattern), so it is genuinely out-of-sample, and the decision logic is shared between the backtest and live inference (`gate_config.py`) so the two cannot drift.

Book	NET Sharpe	vs baseline
baseline (regime-unaware)	+0.372	—
gated blend (uniform global gate)	+0.298	-0.074
per-spread gate (Phase 8)	+0.374	+0.002

Verdict (graded): per-spread gating closes the gap. It lifts the gated/regime book from +0.298 (uniform global gate) to +0.374 — reaching parity with the regime-unaware baseline +0.372. The final-cutoff config enables the regime leg on exactly **wti_fly_123**, **wti_m1_m2** (the spreads whose regime leg earned it OOS) and routes every other spread to the rolling-z baseline. It does **not** beat baseline (consistent with the whole Phase 2.8.x arc) — the value is that it finally makes the regime book *competitive*, deploying regime conditioning only where the per-spread lift table justifies it, and it is a clean ++0.076 over the production global gate. The

verdict is insensitive to the margin / min-sample knobs (NET Sharpe 0.374–0.376 across margin 0–0.25, min_n 10–30), so it is not a fitted edge. Live: on by default (PULSE_PERSPREAD_GATE=0 reverts to the uniform gate).

GARCH conditional vol — does it help the risk layer? (2026-06-22)

Phase 7 vol-targeting sizes each position by a *trailing 20-day realised* \$/bbl vol. This study asks whether a **GARCH conditional-vol forecast** (vol clustering — shocks beget shocks; plain GARCH(1,1) and asymmetric GJR-GARCH, Student-t, fit causally on daily \$/bbl changes) does better. Two questions, two answers.

(1) As a 1-step forecast — no. On QLIKE (lower=better), the trailing-20d window wins on the mean and on 3-of-6 spreads; GARCH/GJR only edge it on the two front spreads. Calendar spreads are low-and-stable-vol, where the clustering edge GARCH exploits on outright prices is muted.

Variant (NET)	realised-20d	plain GARCH	GJR-GARCH
decorrelated	+0.251 / -215 / +1.92	+0.251 / -215 / +1.92	+0.251 / -215 / +1.92
risk_parity	+0.235 / -141 / +1.87	+0.331 / -115 / +2.94	+0.343 / -108 / +3.25
parity_stress	+0.203 / -141 / +1.52	+0.291 / -119 / +2.41	+0.298 / -116 / +2.54
vol_target	+0.198 / -112 / +2.01	+0.277 / -96 / +3.03	+0.285 / -93 / +3.21

Each cell: NET Sharpe / max-DD / Calmar. 'decorrelated' has no vol scaling → identical across inputs (sanity check).

(2) As the sizing input — yes, materially. Feeding GARCH vol into the full vol-target overlay lifts the book from realised-vol +0.198 Sharpe / Calmar +2.01 to GJR-GARCH +0.285 / Calmar +3.21 (max-DD -112 → -93), **robust across plain and GJR**. This turns Phase 7 from a Sharpe-losing drawdown tool into a Calmar-improving one — vol-targeting doesn't need the best point forecast, it needs reactivity to vol onset (GARCH leads the lagging window) and sensible cross-sectional risk. **Verdict (graded):** a genuine **risk-layer refinement, not alpha** — it still sits below the baseline headline (+0.372 Sharpe / Calmar 4.99), so it does not change the conclusion; but it's the first lever that makes the vol-targeted regime book competitive on risk-adjusted terms. **Caveat:** the asymmetric (GJR) edge is small over plain GARCH, and the win is concentrated in risk-parity's cross-sectional reweighting.

Caveats & next steps

- **WTI is synthesised** — last-1-min midprice per session, not exchange print. Mentor's real WTI C1-C3 daily file slots in without code changes.
- **Sparse cells (composite only)** — 96/162 composite cells (BACK × HIGH, CONTANGO × LOW) stay empty. The pooled grid is dense by construction (3 cells/spread, average ~770 rows/cell vs ~150 composite).
- **Phase 1 not directly comparable** — Phase 1 is a 9-indicator directional Brent signal; the baseline here (regime-unaware 252d z on spreads) is the cleanest test of regime-conditioning value.
- **Costs modelled (Phase 2.8.6)** — defensible per-spread RT cost (\$0.030 M1-M2 / \$0.040 M3-M6 / \$0.050 fly per bbl) layered on top of gross spread-move PnL. Roll slippage and funding still ignored; live deployment should add a per-spread broker quote where available.
- **Mode switch** — set PULSE_REGIME_MODE=pooled for un-gated pooled inference, PULSE_GATED_BLEND=1 (Phase 2.6) for the gated blend, and PULSE_GATED_SIZE=<full|half|kelly> (Phase 2.7) to scale the regime-leg notional. Default is composite + full sizing for back-compat with Sprint 3 dashboard cards.
- **Gate is conservative by design** — Phase 2.5 demonstrated lift only on (BACK regime × {Lasso, Huber} winners). The gated blend codifies exactly that slice; per-spread Sharpe and the regime/baseline split in the headline disclose how often the engine actually fires versus deferring to the baseline.
- **Phase 2.7 sizing has a Sharpe-vs-DD trade-off** — uniform 0.5× halves both mean PnL and max DD on the regime leg; Kelly is per-spread and adaptive but its sample is thin (97 regime fires total). Live deployment should read sized_blend_summary.kelly_per_spread_latest from the walk-forward report — the same numbers shown above — to size at inference time.
- **Phase 5 delivered:** the 5/10/20/30d horizon sweep (baseline edge lives at 30d, the regime model's at 5-10d) and Lasso stability-selection feature pruning (leaner but trails full feats on NET Sharpe). **Phase 6 delivered (above):** data-driven HMM / change-point regimes replacing the hard -\$2/+\$5 curve thresholds — the detector relocates the contango cut toward zero but does not unseat the headline, confirming the regime partition is not the binding constraint. **Phase 7 delivered (above):** portfolio vol-targeting halves the gated book's max-drawdown but trades away NET Sharpe/Calmar — a drawdown-management tool, not alpha. **Phase 8 delivered (above):** the per-spread gate replaces the uniform global gate with a per-spread enable decision (regime leg fires only where its OOS NET Sharpe beat baseline) — it lifts the gated book to baseline parity (+0.298 → +0.374) by enabling regime on the two WTI front spreads and routing the rest to baseline. **Phase 2.8.x model backlog is now complete; baseline remains the NET-Sharpe headline across every variant, with the per-spread-gated regime book now matching it.** Open route: reading the live A/B verdict once ≥30 closed trades/arm accumulate.